

MOVING ON FROM GEOMETRY

Lines and the polygons that they delineate are a much less convenient giscience abstraction than simple point locations. However, unlike point locations, polygons in a GIS or on a map directly represent things in the world. In fact, they often do much more than represent a thing in the world, they actually *are* that thing, legally and politically. A line in a geospatial database often has precedence over any physical manifestation of the corresponding boundary on the ground. If a property owner builds a fence that lays claim to more land than is associated with that parcel in a cadastral database, the line on the map is likely to take precedence over the fence in any legal dispute.

This inversion of the map-territory relationship—the map *is* the territory, it *makes* the territory—is to a large extent accepted in both geographical theory and in giscience. Geographers recognize that maps make places, and it is recognized in giscience that many (if not all) lines on maps are fiat objects, that is, “acts of human decision” (Smith, 2001, p. 133). An alternative approach to the fundamentals of geospatial computing follows from the view that fiat objects are the proper geoatoms on which geospatial data structures should be built. This approach is grounded in *mereology*, the philosophical study of part-whole relations, the topology of part-whole relations *mereotopology*, and mathematical treatments of these (Simons, 1987). In giscience discussion of how naïve geography concepts (Egenhofer & Mark, 1995), such as near, far, in front of, around, and so on, might be represented and reasoned with computationally is where these ideas have seen most uptake and interest. An alternative to standard GIS’s foundational 9-intersection model for spatial relations between topological point sets (see Figure 5.1) is provided by *qualitative spatial reasoning* (Cohn & Renz, 2008).

For example, Worboys and Duckham (2021) show how qualitative reasoning about the relations among the Voronoi regions associated with spatial entities (see §The Voronoi Model of Space, Chapter 2) might enable automated descriptions of complex spatial arrangements more meaningful than those offered by the 9-intersection model. Stell discusses these ideas in relation to how space as it is experienced might be

represented computationally, arguing that “qualitative relations can be used computationally as abstractions from quantitative data instead of being seen as an alternative and separate representation” (2017, para. 7). We touch on these ideas again in discussing time and process (see Chapters 7 and 8). Stell (2017) sets out the relevance of these discussions to the present context in discussing how boundaries cannot be coherently handled by point set topologies. Where two spatial entities touch, if our geoatom is an infinitesimal point, then it is impossible to say to which of the two entities a point arbitrarily close to the boundary belongs. This might sound like (literal) philosophical hairsplitting, but it also arises in practical geospatial work in dealing with spatial relations between geometries (very) near one another as a result of imprecision in floating point calculations. Often, the only workaround is to enforce arbitrary precision on calculations, which in effect makes the geoatom not a point, but a tiny pixel and introduces myriad other inconsistencies.

In the present context of dealing with lines, boundaries, polygons, and their relations, such problems reflect the dual nature of boundaries, which contain *and* connect areas (see Figure 5.3). This highlights (again) the importance to any coherent understanding of geographical space of relations, which are the focus of the next chapter.